CoderAxo AI/ML Internship Program — 2026

PROJECT REPORT

Real-Time Driver Drowsiness Detection System

DrowsyGuard

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1. Introduction

Driver drowsiness is a critical road safety issue that leads to thousands of preventable fatalities every year. Unlike distracted driving or speeding, drowsiness is an **internal physiological state** that is difficult for drivers themselves to recognize in time. Traditional safety mechanisms — road rumble strips, lane departure warning systems — are purely reactive and engage only after the vehicle has already drifted into danger.

This project, **DrowsyGuard**, addresses this gap by building a **proactive, real-time drowsiness detection system** that continuously monitors the driver's facial cues using a standard webcam and computer vision algorithms. The system provides timely, multi-modal alerts (audio and visual) before a safety incident can occur.

The project is implemented using **Python**, **OpenCV**, **MediaPipe FaceMesh**, **scikit-learn (SVM)**, **FastAPI**, and **Streamlit**. It covers the complete AI/ML pipeline: data generation, feature engineering, model development, hyperparameter tuning, evaluation, and deployment.

2. Problem Statement

Fatigue and drowsiness are responsible for an estimated **20–30% of all road traffic accidents** globally (WHO, 2023). Studies indicate:

- Drowsy drivers have reaction times comparable to drunk drivers (BAC 0.08%)
- Microsleep episodes (involuntary sleep lasting 1–30 seconds) can occur without the driver's awareness
- Commercial truck and bus drivers are at highest risk due to long, monotonous driving schedules
- Existing OEM in-car solutions (e.g., Mercedes Attention Assist, Volvo DAMS) are expensive and proprietary

There is a clear need for an **affordable, open-source, hardware-agnostic solution** that works on commodity hardware (laptop/dashcam) and can be extended by researchers and developers. DrowsyGuard fulfills this need.

3. Objectives

- Detect driver drowsiness in real-time (< 500ms latency) using facial landmark analysis
- Implement Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and Head Pose estimation algorithms
- Train a supervised ML classifier (SVM) on extracted features achieving ≥ 90% accuracy
- Classify driver states into three levels: Alert → Drowsy → Very Drowsy

• Design a two-tier alert escalation system (soft alert at 2s, hard alert at 4s)
• Deploy a live monitoring dashboard using Streamlit
• Log all drowsiness events for post-session review and analysis

4. Dataset Description

Two public benchmark datasets were referenced for this project, along with a synthetic feature dataset generated from the feature extraction pipeline:

Dataset	Source	Size	Classes	Usage
YawDD (Yawning Detection)	Kaggle	~800 MB	Yawning / Non-Yawning	Video analysis reference
NTHU-DDD (Drowsy Driver)	Kaggle	~2.1 GB	Alert / Drowsy / Sleepy	Validation benchmark
Synthetic EAR/MAR Dataset	Generated (this project)	5000 samples	Alert / Drowsy / Very Drowsy	Model training

4.1 Synthetic Dataset Generation

The synthetic data was generated using statistical distributions derived from published benchmark studies:

Class	Label	Samples	EAR Range	MAR Range	PERCLOS Range
Alert	0	2,500	0.22–0.45	0.10–0.60	0.00–0.20
Drowsy	1	1,500	0.12–0.26	0.35–0.85	0.15–0.60
Very Drowsy	2	1,000	0.05–0.20	0.60–1.20	0.40–0.95

Features: **EAR**, **MAR**, **head_pitch**, **PERCLOS**, **ear_mar_ratio**, **pitch_norm**, **drowsy_score**.

5. Methodology

5.1 System Pipeline

Layer	Component	Technology	Function
1	Video Capture	OpenCV VideoCapture	Acquire frames at 20–30 FPS
2	Face Detection	MediaPipe FaceMesh	Locate face, 468 landmarks
3	Feature Extraction	Custom Python (scipy)	Compute EAR, MAR, Head Pose
4	State Classifier	SVM (scikit-learn)	Predict: Alert/Drowsy/Very Drowsy
5	Alert Engine	pygame + OpenCV	Audio beep + visual overlay

6	API Server	FastAPI + WebSocket	Stream state data to frontend
7	Dashboard	Streamlit	Live monitoring interface
8	Event Logger	Python CSV writer	Persist drowsiness events

5.2 Data Preprocessing

- Outlier removal using IQR clipping (EAR $\in$ [0, 0.5])
- Feature engineering: ear_mar_ratio, pitch_norm, drowsy_score composite feature
- Stratified 70/10/20 train/validation/test split
- StandardScaler normalization fitted only on training data

5.3 Alert Escalation Logic

Time (s)	Condition	Alert Level	Action
0	EAR drops below 0.22	Detection start	Frame counter begins
2	EAR below threshold (2s)	SOFT ALERT	Beep sound (1000 Hz, 200ms)
4	EAR below threshold (4s)	HARD ALERT	Loud alarm + red screen flash
Reset	EAR recovers for 3 frames	ALERT restored	Counter resets, alerts stop

6. Algorithms Used

6.1 Eye Aspect Ratio (EAR)

The EAR algorithm was introduced by Soukupová and ■ech (2016). It quantifies eye openness using six facial landmarks:

```
EAR = (||p2 - p6|| + ||p3 - p5||) / (2.0 × ||p1 - p4||)
```

When both eyes are open, EAR ≈ 0.25–0.35. A sustained EAR below **0.22** for more than 2 seconds indicates drowsiness.

6.2 Mouth Aspect Ratio (MAR)

MAR detects yawning using 12 mouth landmark coordinates:

```
MAR = (||p2-p8|| + ||p3-p7|| + ||p4-p6||) / (2.0 × ||p1-p5||)
```

Normal: MAR < 0.5. Yawning: MAR > **0.65**.

6.3 Head Pose Estimation

Head pitch is estimated using OpenCV's solvePnP with six 3D reference landmarks:

```
cv2.solvePnP(model_3d_points, image_2d_points, camera_matrix, dist_coeffs)
```

```
pitch = math.degrees(math.asin(-rotation_matrix[2][0]))
```

A pitch exceeding **20°** combined with low EAR indicates microsleep.

6.4 PERCLOS

PERCLOS (Wierwille & Ellsworth, 1994) measures eye closure percentage over a rolling 60-frame window:

```
PERCLOS = count(EAR[i] < EAR_THRESHOLD) / 60 frames
```

- PERCLOS < 0.20 → Alert
- 0.20–0.50 → Drowsy
- > 0.50 → Very Drowsy

6.5 SVM Classifier

A **Support Vector Machine** with RBF kernel is selected because: the EAR/MAR feature space has non-linear class boundaries; SVMs are fast at inference (real-time critical); and they generalize well with limited data. Best hyperparameters (GridSearchCV, 5-fold CV): **C=10, gamma='scale', kernel='rbf'**.

7. Experimental Results

Model	Accuracy	Precision	Recall	F1-Score	Train Time
Rule-Based (Baseline)	85.3%	84.7%	85.3%	84.9%	<1s
SVM (RBF)	93.8%	93.5%	93.8%	93.6%	1.2s
Random Forest	94.0%	93.8%	94.0%	93.8%	3.5s
Gradient Boosting	94.4%	94.1%	94.4%	94.2%	8.7s
SVM (Tuned, C=10)	94.2%	93.9%	94.2%	94.0%	2.1s

The **SVM (Tuned)** is selected for deployment — optimal balance of accuracy and inference speed.

7.1 Per-Class Performance — Best SVM

Class	Precision	Recall	F1-Score	Support
Alert	95.1%	96.3%	95.7%	500
Drowsy	93.4%	92.8%	93.1%	300
Very Drowsy	94.2%	93.5%	93.8%	200
Macro Avg	94.2%	94.2%	94.2%	1000

7.2 AUC-ROC Scores

Class	AUC-ROC	Interpretation
Alert	0.989	Excellent — very high separation from non-alert
Drowsy	0.971	Excellent — reliable detection of fatigue onset
Very Drowsy	0.983	Excellent — near-perfect critical state detection
Macro Average	0.981	Excellent overall discriminative ability

8. Performance Evaluation

Metric	Target	Achieved	Status
Detection Accuracy	≥ 90%	94.2%	✓ PASS
AUC-ROC	≥ 0.95	0.981	✓ PASS
Inference Latency	< 500ms	~40ms	✓ PASS
FPS (i5 CPU)	≥ 20	24–28	✓ PASS
Alert Trigger Time	< 2.5s	2.0s	✓ PASS
False Positive Rate	< 5%	3.8%	✓ PASS
Dashboard Load Time	< 3s	~1.5s	✓ PASS

The system processes each frame in approximately **35–45ms** on a standard Intel Core i5 (8GB RAM, no GPU), achieving 24–28 FPS sustained.

9. Challenges Faced

• Occlusion & Glasses

MediaPipe landmark detection degrades with sunglasses. Mitigated with a face confidence threshold and head-pose-only fallback.

• Variable Lighting

EAR values shift under low-light. Addressed with CLAHE preprocessing.

• False Positives from Natural Blinks

Normal blinks triggered alerts. Solved by requiring EAR below threshold for ≥ 48 consecutive frames (2 seconds at 24 FPS).

• Head Pose Instability

solvePnP was noisy on fast movement. Applied EMA: pitch_smooth = 0.7 × pitch_prev + 0.3 × pitch_raw.

• Real-Time Dashboard Latency

WebSocket lag at high resolution. Fixed by compressing JPEG to 70% and resizing to 640×480.

• Class Imbalance

Very Drowsy class underrepresented. Addressed with stratified split and class_weight='balanced'.

10. Conclusion

DrowsyGuard successfully demonstrates that **real-time driver drowsiness detection is achievable with commodity hardware and open-source tools**. By combining EAR, MAR, and head pitch with a tuned SVM, the system achieves **94.2% accuracy** with sub-50ms latency.

The modular architecture allows independent improvement of any layer. The Streamlit deployment makes it immediately accessible for research and demonstration. The project fulfills all CoderAxo internship requirements across the complete AI/ML pipeline.

11. Future Scope

- BiLSTM Temporal Model: Process 60-frame EAR sequences for temporal pattern detection
- Night Vision: IR/thermal camera preprocessing for nighttime reliability
- YOLOv8-Face: Superior side-face and occlusion handling
- Mobile App (Flutter + TFLite): On-device dashcam integration
- Physiological Fusion: Combine EAR with remote PPG heart rate
- Fleet Monitoring: Scale to 8+ streams with Redis + cloud dashboard
- Driver Identification: Personalize EAR thresholds per driver via face recognition

12. References

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[5] Cortes, C., & Vapnik, V. (1995). Support-vector networks. Machine Learning, 20(3), 273–297.

[6] YawDD Dataset: https://www.kaggle.com/datasets/davidvazquezcic/yawn-dataset

[7] NTHU-DDD Dataset: https://www.kaggle.com/datasets/rakibuleceruet/drowsiness-prediction-dataset

[8] WHO Road Traffic Injuries Fact Sheet (2023). World Health Organization.

[9] FastAPI Documentation: https://fastapi.tiangolo.com

[10] Streamlit Documentation: https://docs.streamlit.io

Project Submission Link

Google Drive Link:

https://drive.google.com/file/d/18OsHAQoYa8i1E8_BlLshkBVnQow0jPuY/view?usp=sharing